Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Development of Design, Machining, and Non-Contact Measurement Technology for 300 mm-Class SiC Free-Form Optics for Space / Semiconductor Applications*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery | materials		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Grinding/Polishing Machine Tools*	*Ceramic Materials*	*Ceramic Manufacturing Process Technologies*
Concept of Proposed Technology	*The core concept is a fully integrated process for the design, ultra-precision machining, and real-time non-contact measurement of 300 mm-class SiC free-form optics with shape precision below 500 nm. This approach leverages lightweight SiC materials, polishing, and in-situ digital metrology to meet the demanding requirements of space systems.*			
Rationale for Joint R&D Project	*This joint R&D project is essential because the advanced technologies required, such as SiC ultra-precision machining, free-form optics fabrication, and in-situ metrology, cannot be developed domestically within a short timeframe. International collaboration will significantly reduce development time and costs, while enabling access to global standards and new markets.*			
Key Words	1. SiC free-form optics 2. Ultra-precision grinding and polishing 3. Non-contact in-situ measurement 4. Digital process chain			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 6
R&D Objectives	Next-generation semiconductor lithography and space-borne optical systems present a growing demand for large-aperture, 300 mm-class free-form reflectors and optical surfaces that achieve shape precision below 500 nm. Meeting these stringent requirements necessitates both material innovation and process advancement. Silicon carbide (SiC) has emerged as a highly promising mirror material for satellites and space applications, primarily due to its outstanding combination of high specific stiffness, offering an excellent ratio of elasticity to density, and its extremely low thermal expansion coefficient, which ensures dimensional stability even under fluctuating environmental conditions in space. However, the manufacture of such high-performance optics faces significant challenges, particularly when working with difficult-to-machine materials like SiC and producing complex free-form geometries. While current micro-polishing methods can achieve wavefront error (WFE) levels around 20 nm RMS for simpler shapes, they are insufficient for consistent quality and process control in advanced applications. As a result, there is a critical need for the development of advanced process and metrology technologies tailored to the unique properties of SiC and the demands of free-form optical surfaces. A central objective of the project is to establish and deploy non-contact surface and shape measurement technologies that can be applied directly during the machining process (on-machine/in-situ). This real-time metrology is essential to ensure that the required shape precision is consistently achieved and maintained throughout fabrication, enabling immediate correction and minimizing rework or scrap. To address these technical challenges, the project leverages an integrated approach that combines design, machining, measurement, and process optimization. This is made possible through a strong international collaboration, especially with Fraunhofer IPT, recognized for its leadership in process-critical components, lightweight system design, and modular optical solutions. By uniting these capabilities, the project aims not only to push the boundaries of what is technically feasible but also to set new standards for the digitalization, efficiency, and reliability of large-aperture, ultra-precision SiC free-form optics manufacturing.			
R&D Target Specifications	The R&D target specifications are focused on the following: - Delivering 300 mm-class reflective SiC free-form optics - Shape accuracy of ≤ 500 nm RMS - Implementation of non-contact, in-situ measurement systems			

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	- Measurement precision also of ≤ 100 nm RMS - Ensuring real-time process control - Fully digitlized process - Optimized for lightweight, high-performance reflective optics suitable for demanding applications in space and semiconductor lithography - Technology transfer and commercialization infrastructure
R&D Plan	The R&D plan is structured in two main phases to ensure systematic development and risk mitigation. Phase 1, lasting 6 months, will focus on a comprehensive pre-study that includes market and technology analysis, a review of existing patents, and detailed benchmarking of current solutions. During this phase, technical specifications and requirements will be defined, collaboration structures and communication protocols will be established, and the necessary infrastructure and key process components will be identified and secured. This phase will conclude with a readiness assessment and a detailed plan for the main development activities for phase II. Phase 2, spanning 30 months, will concentrate on the development and optimization of design and simulation methods for SiC free-form optics, the construction and digitalization of the manufacturing process chain, and the development of specialized grinding tools and robotic equipment. Key activities will include the prototyping of both robotic polishing and non-contact, in-situ measurement systems, as well as iterative testing and validation to achieve the target precision on real demonstrators. This phase also covers the implementation of safety and security protocols, preparation for technology transfer and commercialization, and extensive documentation and training to facilitate the adoption and scaling of the developed technologies.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Advanced Research The research institute will assume a central role in driving scientific innovation and technical progress throughout the project. This includes the design and simulation of free-form SiC optics, which demand specialized expertise in optical engineering and computational modeling. IPT will use advanced ultra-precision machining methods and metrology technologies, ensuring that all process steps, from initial concept to final product, meet the stringent requirements of space and semiconductor applications. By developing and digitalizing the entire process chain, the research institute will link design, machining, and measurement in a seamless workflow, enabling enhanced process control, reproducibility, and efficiency. Comprehensive process and quality analyses will be performed to optimize each stage and mitigate potential risks, thereby guaranteeing consistently high product quality. Metrology A further key responsibility of the research institute lies in establishing robust measurement protocols for SiC and free-form optics manufacturing. The institute will look for advanced metrology solutions capable of verifying the shape accuracy and surface quality of complex optical components, which is essential for industrial acceptance and market access. By collaborating with Korean Partners, the institute will help to define benchmarks for manufacturing and verification, thereby facilitating the commercialization and use of the developed technologies. These efforts will also contribute to reducing overseas dependency and strengthening domestic capabilities in ultra-precision optics. Transfer of knowledge to Korean industry A crucial aspect of the research institute's role is the transfer of knowledge to Korean industry partners, achieved through close international collaboration and expertise exchange. By leveraging Fraunhofer IPT proven competence in free-form optics design and manufacturing, the institute will help Korean SMEs rapidly acquire core technologies and best practices. Joint activities such as workshops, technical training, and collaborative development projects will foster the dissemination of innovative techniques and lessons learned from process modeling and experimental validation. This approach accelerates prototype development, shortens production chain establishment timelines, and promotes the adoption of advanced manufacturing processes within Korea's industrial ecosystem. Ultimately, effective knowledge transfer will drive innovation, improve production efficiency, and ensure the long-term sustainability and competitiveness of the project outcomes on a global scale.
Expected R&R of Korean SME Partner	Project management Project management will oversee the entire duration of the project, ensuring that all work packages are executed efficiently and within the established timeline. This structured approach is essential for coordinating activities, managing resources, and mitigating risks, thereby facilitating the successful completion of project objectives. Prototyping The Korean SME partner will focus on industrial implementation, including prototyping, process adaptation, and scaling up the developed technologies for production. They will

	establish and operate the necessary manufacturing and measurement infrastructure, conduct pilot and mass production runs, and optimize workflows based on project outcomes. The SME will lead market introduction, customer engagement, and support commercialization efforts, ensuring that the developed solutions are effectively transferred into real-world applications and industry adoption is accelerated. <u>Scalability</u> The Korean SME partner will be responsible for integrating the developed technologies into prototype and production environments, including adapting processes for local manufacturing needs and refining them for scalability. They will set up and operate the required production and metrology infrastructure, carry out testing and validation of prototypes, and drive the transition to mass production. Additionally, the SME will handle market introduction, customer engagement, and regulatory compliance, ensuring effective commercialization and industry uptake of the new SiC free-form optics solutions.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project is expected to drive the digitalization and advancement of SiC ultra-precision free-form optical manufacturing processes, enabling higher efficiency, reproducibility, and process optimization. By developing and implementing lightweight free-form optical systems, the project will provide cutting-edge solutions for space-borne telescopes, enhancing the performance and payload capabilities of next-generation space missions. The integration of non-contact, in-situ measurement and shape correction technologies will ensure real-time quality assurance and enable ultra-precision manufacturing at industrial scale. These innovations will reduce overseas dependency and help stabilize the supply chain within the optical components and equipment sector, strengthening domestic manufacturing resilience. Furthermore, by taking a leading role in the standardization of ultra-precision SiC and free-form optics manufacturing for space and semiconductor applications, the project will establish international benchmarks and facilitate global market access. Finally, the outcomes are expected to significantly strengthen the commercialization and export foundation for related fields such as free-form optics manufacturing for space and semiconductor applications, driving broader industrial innovation and economic growth.

Klicken oder tippen Sie hier, um Text einzugeben.☐ **Terms:**

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization